

Untitled16

October 11, 2025

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[2]: from google.colab import drive
drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

```
[3]: import numpy as np
filename = '/content/drive/MyDrive/DSC257R/glove.6B/glove.6B.300d.txt'
with open(filename) as f:
    content = f.read().splitlines()
n = len(content)
vecs = np.zeros((n,300))
words = []
index = 0
for rawline in content:
    line = rawline.split()
    words.append(line[0])
    vecs[index] = np.genfromtxt(line[1:])
    index = index+1
```

```
[8]: word_list = ['communism', 'africa', 'happy', 'sad', 'upset', 'computer', 'cat', '
↪ 'dollar']
```

```
[4]: from sklearn.neighbors import NearestNeighbors

def get_nearest_neighbors(x_train, x_test, n_neighbors=5):
    # create NearestNeighbors model
    nn_model = NearestNeighbors(n_neighbors=n_neighbors, algorithm='auto')

    # fit the model on training data
    nn_model.fit(x_train)

    # find the nearest neighbors for each test sample
    distances, indices = nn_model.kneighbors(x_test)

    return distances, indices
```

```
[13]: closest_five_words = {}
      for w in word_list:
          idx = words.index(w)
          v = vecs[idx].reshape(1,-1)
          dis, idx = get_nearest_neighbors(vecs, v, 6)
          print(f"Five closest words to {w}:")
          closest_five_words[w] = []
          for i in range(1,6):
              closest_five_words[w].append(words[idx[0][i]])
          print(f"{words[idx[0][i]]}")
```

Five closest words to communism:

fascism
capitalism
nazism
stalinism
socialism

Five closest words to africa:

african
continent
south
africans
zimbabwe

Five closest words to happy:

glad
pleased
always
everyone
sure

Five closest words to sad:

sorry
tragic
happy
pathetic
awful

Five closest words to upset:

upsetting
surprised
upsets
stunned
shocked

Five closest words to computer:

computers
software
technology
laptop
computing

Five closest words to cat:

cats

dog

pet

feline

dogs

Five closest words to dollar:

currency

dollars

euro

multibillion

weaker